

Foundations of Cybersecurity

Module 1

Welcome to the exciting world of cybersecurity

Module 2

The evolution of cybersecurity

The history of cybersecurity

Welcome to module 2

Video • 3 min

First cybersecurity attacks

Video • 3 min

Attacks in the digital age

Video • 3 min

Common attacks and their effectiveness

Reading • 12 min

Identify methods of attack

Ungraded Quiz • 10 min

Scan: Keep your cool during a data breach

Video • 2 min

Test your knowledge: The history of cybersecurity

Practice Assignment • Grade: 100%

The eight CISAP security domains

Introduction to the eight CISAP security domains, Part 1

Video • 4 min

Introduction to the eight CISAP security domains, Part 2

Video • 3 min

Intersect the type of attack

Reading • 4 min

Understand attacks

Reading • 3 min

Test your knowledge: The eight CISAP security domains

Practice Assignment • Grade: 100%

Review: The evolution of cybersecurity

Wrap-up

Video • 1 min

Quizzary items from module 2

Reading • 8 min

Module 2 challenge

Ungraded Assignment • Grade: 100%

Module 3

Protect against threats, risks, and vulnerabilities

Frameworks and controls

Welcome to module 3

Video • 2 min

Introduction to security frameworks and controls

Video • 4 min

Secure design

Video • 3 min

Controls, frameworks, and compliance

Reading • 12 min

Interact: Protect sensitive data and information

Video • 2 min

Test your knowledge: Frameworks and controls

Practice Assignment • Grade: 100%

Ethics in cybersecurity

Ethics in cybersecurity

Video • 4 min

Ethical concepts that guide cybersecurity decisions

Reading • 12 min

Practice Ethics in cybersecurity professionals

Ungraded Quiz • 10 min

Ready: The importance of ethics as a cybersecurity professional

Video • 2 min

Test your knowledge: Ethics in cybersecurity

Practice Assignment • Grade: 100%

Review: Protect against threats, risks, and vulnerabilities

Wrap-up

Video • 1 min

Quizzary items from module 3

Reading • 4 min

Module 3 challenge

Ungraded Assignment • Grade: 100%

Module 4

Cybersecurity tools and programming languages

Important cybersecurity tools

Welcome to module 4

Video • 1 min

Overview cybersecurity tools

Video • 4 min

Tools for protecting business operations

Reading • 12 min

Explore: Tools and their purposes

Ungraded Quiz • 10 min

Test your knowledge: Important cybersecurity tools

Practice Assignment • 8 min

Core cybersecurity knowledge and skills

Introduction to Linux, SQL, and Python

Video • 3 min

Use tools to protect business operations

Reading • 12 min

Test your knowledge: Core cybersecurity knowledge and skills

Practice Assignment • 8 min

Create a cybersecurity portfolio

Reading • 12 min

Portfolio Activity: Craft a professional statement

Ungraded Assignment • 20 min

Portfolio Activity: Compare: Craft a professional statement

Reading • 4 min

Review: Cybersecurity tools and programming languages

Coach Dialogue: Explore entry-level security analyst responsibilities

Ungraded Quiz • 10 min

Wrap-up

Video • 42 sec

Quizzary items from module 4

Reading • 2 min

Module 4 challenge

Ungraded Assignment • 40 min

Congratulations on completing Course 1!

Join the community

Reading • 4 min

Course wrap-up

Video • 1 min

Course 1 gallery

Reading • 1 min

Get started on the next course

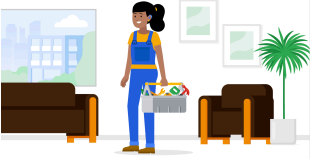
Reading • 4 min

Tools for protecting business operations

Previously, you were introduced to several technical skills that security analysts need to develop. You were also introduced to some tools entry-level security analysts may have in their toolkit. In this reading, you'll learn more about how technical skills and tools help security analysts mitigate risks.

An entry-level analyst's toolkit

Every organization may provide a different toolkit, depending on its security needs. As a future analyst, it's important that you are familiar with industry standard tools and can demonstrate your ability to learn how to use similar tools in a potential workplace.



Security information and event management (SIEM) tools

A **SIEM tool** is an application that collects and analyzes log data to monitor critical activities in an organization. A **log** is a record of events that occur within an organization's systems. Depending on the amount of data you're working with, it could take hours or days to filter through log data on your own. SIEM tools reduce the amount of data an analyst must review by providing alerts for specific types of threats, risks, and vulnerabilities.

SIEM tools provide a series of dashboards that visually organize data into categories, allowing users to select the data they wish to analyze. Different SIEM tools have different dashboard types that display the information you have access to.

SIEM tools also come with different hosting options, including on-premise and cloud. Organizations may choose one hosting option over another based on a security team member's expertise. For example, because a cloud-hosted version tends to be easier to set up, use, and maintain than an on-premise version, a less-experienced security team may choose this option for their organization.

Network protocol analyzers (packet sniffers)

A **network protocol analyzer** (also known as a **packet sniffer**), is a tool designed to capture and analyze data traffic in a network. This means that the tool keeps a record of all the data that a computer within an organization's network encounters. Later in the program, you'll have an opportunity to practice using some common network protocol analyzer (packet sniffer) tools.

Playbooks

A **playbook** is a manual that provides details about any operational action, such as how to respond to a security incident. Organizations usually have multiple playbooks documenting processes and procedures for their teams to follow. Playbooks vary from one organization to the next, but they all have a similar purpose: To guide analysts through a series of steps to complete specific security-related tasks.

For example, consider the following scenario: You are working as a security analyst for an incident response firm. You are given a case involving a small medical practice that has suffered a security breach. Your job is to help with the forensic investigation and provide evidence to a cybersecurity insurance company. They will then use your investigative findings to determine whether the medical practice will receive their insurance payout.

In this scenario, playbooks would outline the specific actions you need to take to conduct the investigation. Playbooks also help ensure that you are following proper protocols and procedures. When working on a forensic case, there are two playbooks you might follow:

- The first type of playbook you might consult is called the **chain of custody** playbook. Chain of custody is the process of documenting evidence possession and control during an incident response. As a security analyst involved in a forensic analysis, you will work with the computer data that was breached. You and the forensic team will also need to document who, what, where, and why you have the collected evidence. The evidence is your responsibility while it is in your possession. Evidence must be kept safe and tracked. Every time evidence is moved, it should be reported. This allows all parties involved to know exactly where the evidence is at all times.
- The second playbook your team might use is called the **protecting and preserving evidence** playbook. Protecting and preserving evidence is the process of properly working with fragile and volatile digital evidence. As a security analyst, understanding what fragile and volatile digital evidence is, along with why there is a procedure, is critical. As you follow this playbook, you will consult the **order of volatility**, which is a sequence outlining the order of data that must be preserved from first to last. It prioritizes volatile data, which is data that may be lost if the device in question powers off, regardless of the reason. While conducting an investigation, improper management of digital evidence can compromise and alter that evidence. When evidence is improperly managed during an investigation, it can no longer be used. For this reason, the first priority in any investigation is to properly preserve the data. You can preserve the data by making copies and conducting your investigation using those copies.

Key takeaways

In this reading, you learned about a few tools a security analyst may have in their toolkit, depending on where they work. You also explored two important types of playbooks: chain of custody and protecting and preserving evidence. However, these are only two procedures that occur at the beginning of a forensic investigation. If forensic investigators interest you, you are encouraged to further explore this career path or security practice. In the process, you may learn about forensic tools that you want to add to your toolkit. While all of the forensic components that make up an investigation will not be covered in this certificate program, some forensic concepts will be discussed in later courses.

Resources for more information

The Google Cybersecurity Action Team's [Threat Response Basics](#) provides strategic intelligence for dealing with threats to cloud enterprise.

The Cybersecurity & Infrastructure Security Agency (CISA) has a list of [Cyber Vulnerability Services and Tools](#).

Review the list to learn more about open-source cybersecurity tools.

Work is completed

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coach



Hi, Venkatesh!
How can I help?

Explain this topic in simple terms

Give me a summary

Give me practice questions

Give me real-life examples

Ask me anything

6

Go to next item

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